Question 1 :

Figure 1: Relationship between risk and loggdp

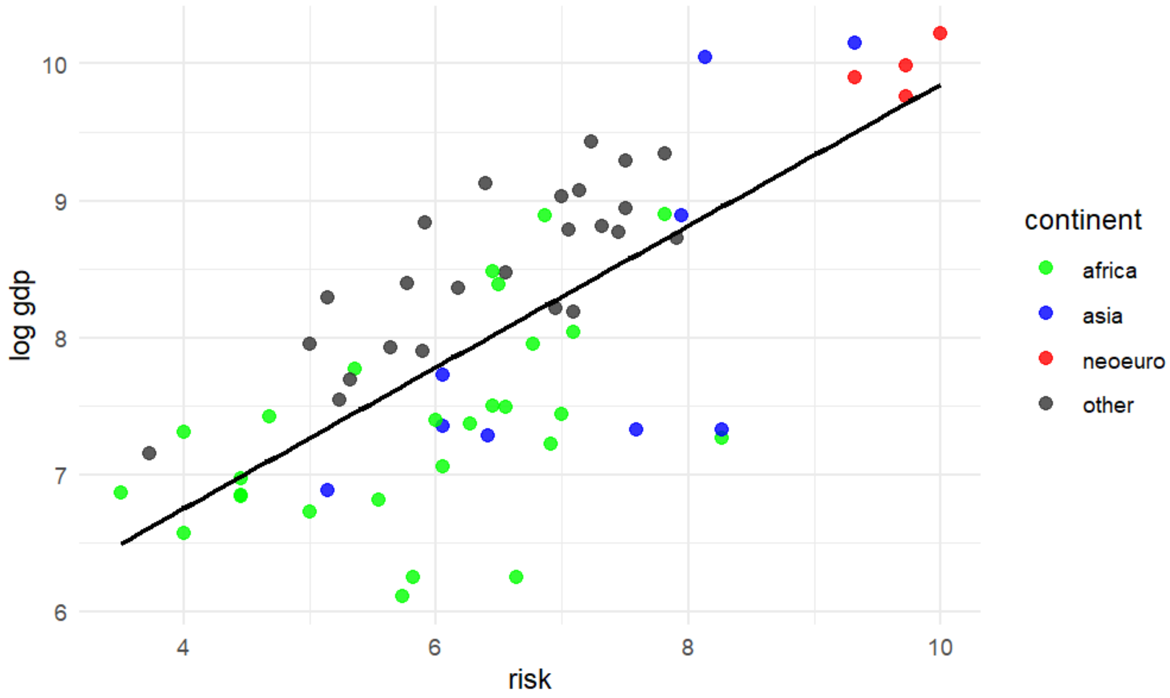

Table 1: OLS regression of log GDP per capita on institutional risk

	Dependent variable:
	Log GDP per capita
Risk	0.516***
	(0.063)
Constant	4.687***
	(0.417)
Observations	64
R^2	0.524
Adjusted R^2	0.516
Residual Std. Error	0.729 (df = 62)
F Statistic	68.170*** (df = 1; 62)
Note:	*p<0.1; **p<0.05; ***p<0.01

We estimate the OLS model : $_i = \alpha + \beta risk_i + u_i$. With an F-statistic of 68.17 ($p < 0.01$), we reject the null hypothesis ($H_0 : \beta_{\text{risk}} = 0$): expropriation protection strongly correlates with loggdp. The R^2 (0.524) indicates that institutional quality is associated with 52.4% of loggdp variation. Assuming a causal relationship, the 0.516 slope implies a 1-point score increase raises loggdp by 0.516, or 68% in levels ($\exp(0.516) \approx 1.68$). Comparing Congo (4.68) and Colombia (7.32), the 2.64-point difference predicts a 1.36 log point gap. Thus, Colombia's predicted GDP is 3.9 times Congo's ($\exp(1.36) \approx 3.91$). Current cross-country data exhibit a similar tend (colombia 2024 gdp per capita is about $21,000 versus $7,000 in congo). This indicates a strong positive association between institutional quality and economic performance. Figure 1 confirms that strong positive correlation. Also, clear regional clustering exists: "Neo-European" nations occupy the high-protection/high-income quadrant, while many African nations concentrate at the lower end. Asian countries exhibit heterogeneity.

Question 2:

Table 2: OLS regression of log GDP per capita on institutional risk

	Dependent variable: Log GDP per capita	
	OLS SE	Robust SE (HC0)
	(1)	(2)
Risk	0.516***	0.516***
	(0.063)	(0.050)
Constant	4.687***	4.687***
	(0.417)	(0.319)
Observations	64	64
R^2	0.524	0.524
Adjusted R^2	0.516	0.516
Residual Std. Error (df = 62)	0.729	0.729
F Statistic (df = 1; 62)	68.170***	68.170***

Note: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$. Column (1) reports OLS standard errors. Column (2) reports heteroskedasticity-robust standard errors (HC0).

Robust SE leaves the coefficient and its significance unchanged. To assess heteroskedasticity, we conduct a Breusch–Pagan test (table 3) with null hypothesis H_0 : $\mathrm{Var}(\varepsilon_i \mid X_i) = \sigma^2$. The test fails to reject homoskedasticity, as the p-value (0.2975) exceeds 5%. Under the classical linear model, the Gauss–Markov theorem implies that OLS is the best linear unbiased estimator, so conventional OLS standard errors are appropriate. However robust standard errors remains useful as a robustness check, since the Breusch-Pagan test provides statistical evidence rather than definitive proof.

Table 3: Breusch–Pagan test for heteroskedasticity

Test	Statistic
Breusch–Pagan $\chi^2(1)$	1.085
df	1
p-value	0.298

Question 3

Table 4: OLS regression of loggdp

	Dependent variable:	
	Log GDP per capita	
	(1)	(2)
Risk	0.516^{***}	0.377^{***}
	(0.063)	(0.061)
latitude		1.382^{**}
		(0.644)
africa		-0.723^{***}
		(0.171)
Constant	4.687^{***}	5.652^{***}
	(0.417)	(0.415)
Observations	64	64
R^2	0.524	0.664
Adjusted R^2	0.516	0.647
Residual Std. Error	0.729 (df = 62)	0.623 (df = 60)
F Statistic	68.170*** (df = 1; 62)	39.472*** (df = 3; 60)
Note:		*p<0.1; **p<0.05; ***p<0.01

Adding latitude and an Africa dummy reduces the residual standard error from 0.729 to 0.623, indicating lower unexplained dispersion. Column (2) estimates

$$loggdp_i = \alpha + \beta_1 risk_i + \beta_2 latitude_i + \beta_3 africa_i + u_i.$$

Latitude and Africa are significantly associated with loggdp at the 5% and 1% levels, conditional on risk. Assuming a causal relationship, the risk coefficient (0.377) implies that a one-point increase in protection raises loggdp by 0.377 (46%), ceteris paribus. The latitude coefficient (1.382) implies that a 0.1 increase (0–1 scale) raises GDP by about

15%. Then, the latitude gap between Zaire (0) and Canada (0.67) predicts a 152% higher GDP.

The Africa dummy (0.723) implies 51% lower GDP for African countries, conditional on controls. Model (2) fits better, with adjusted R^2 rising to 0.647 and residual error falling. The decline in the risk coefficient from 0.516 to 0.377 indicates positive omitted-variable bias in column (1), while precision improves as its SE falls to 0.061.

Question 4 Institutions is potentially endogenous because :

Reverse Causality: While institutions impact GDP, wealth also improves institutions. Richer nations can better fund legal systems and enforcement, while larger tax bases reduce government reliance on predatory policies.

Omitted Variables: Determinants of long-run income (religion, ethnic fractionalization, human capital, or resource dependence) often correlate with institutional quality. Omitting these causes bias, as OLS captures these external effects rather than pure causality.

Ex-post Construction & Perception Bias: Institutional indices are subjective expert assessments. Unlike objective data like latitude, these scores may suffer from cognitive bias, where coders perceive richer, stable countries as having "better" institutions. This creates a mechanical correlation which reinforces endogeneity. Moroever, endogeneity is exacerbated by attenuation bias from measurement error in the institutional index, which is only a proxy for a broader cluster of institutions, and by selection bias, as OLS compares structurally different countries.

Implications: Consequently, the OLS coefficient is inconsistent. Causal interpretations are likely biased by endogeneity, potentially leading to misleading policy conclusions and misallocated resources if development strategies rely on this model.

Question 5

Exogeneity: Settler mortality is assumed uncorrelated with the error term, implying historical mortality was not driven by unobserved determinants of current development. Critics argue that geography or climate could explain both high mortality and low income. The authors counter that malaria and yellow fever mainly affected Europeans, while indigenous populations had acquired immunity, so mortality does not proxy for a general disease environment. Settlement choices were driven by mortality risk rather than initial economic potential. Results are robust to controls for current disease.

Relevance: Mortality shaped settlement patterns: high mortality led to extractive institutions, while low mortality encouraged "Neo-Europes" with strong property rights. Results show a strong correlation between historical mortality and modern institutional quality.

Exclusion Restriction: Settler mortality affects current GDP only through institutions. Conditional on institutions, centuries-old mortality has no plausible direct effect on modern income.

The logarithm of mortality is preferred because : proportional changes are more relevant than levels, for which there is no clear theoretical justification. Logs reduce the influence of extreme values(eg.Africa), limit heteroskedasticity, has an intuitive interpretation and yield a more stable, approximately linear relationship with institutions.

Estimations

Table 5: first stage: determinants of institutions (risk)

	risk
log settler mortality	-0.441**
	(0.170)
latitude	2.040
	(1.334)
africa	-0.317
	(0.394)
Constant	8.329***
	(0.858)
Observations	64
R^2	0.307
Adjusted R^2	0.272
Residual Std. Error	1.253 (df = 60)
F Statistic	8.865*** (df = 3; 60)
Note:	*p<0.1; **p<0.05; ***p<0.01 robust standard errors in parentheses

The first stage shows that the instrument significantly predicts risk ($p = 0.012$).

Table 6: reduced form: effect of settler mortality on income

	log(gdp)
log settler mortality	-0.353***
	(0.098)
latitude	1.577**
	(0.770)
africa	-0.601**
	(0.228)
Constant	9.658***
	(0.495)
Observations	64
R^2	0.546
Adjusted R^2	0.524
Residual Std. Error	0.723 (df = 60)
F Statistic	24.101*** (df = 3; 60)
Note:	*p<0.1; **p<0.05; ***p<0.01
	robust standard errors in parentheses

The reduced form, which directly relates settler mortality to gdp, shows a negative association between log settler mortality and loggdp.

Table 7: dependent variable: log(gdp)

	OLS	*instrumental variable*
	ols	2sls
	(1)	(2)
risk	0.377***	0.800***
	(0.061)	(0.258)
latitude	1.382**	-0.055
	(0.644)	(1.200)
africa	-0.723***	-0.348
	(0.171)	(0.316)
Constant	5.652***	2.995*
	(0.415)	(1.633)
Observations	64	64
R^2	0.664	0.392
Adjusted R^2	0.647	0.362
Residual Std. Error (df = 60)	0.623	0.837
F Statistic	39.472*** (df = 3; 60)	
Note:	*p<0.1; **p<0.05; ***p<0.01 robust standard errors in parentheses	

In the second stage (2SLS), the coefficient on risk is significant. The slope (0.800) exceeds the OLS estimate (0.377), as OLS was likely biased downward by measurement error or endogeneity. In 2SLS, latitude and the Africa dummy become insignificant, suggesting no robust independent effect once institutions are instrumented, albeit with less precise estimates (0.258 vs 0.061 in ols).

Table 8: instrumental variables estimation (2sls)

	dependent variable: log(gdp)
risk	0.800^{***}
	(0.258)
latitude	-0.055
	(1.200)
africa	-0.348
	(0.316)
constant	2.995^{*}
	(1.633)
observations	64
r^2	0.392
adjusted r^2	0.362
residual se	0.837
wald test (p-value)	1.93×10^{-8}
diagnostic tests	
weak instruments f-statistic	6.708
wu–hausman p-value	0.0186
sargan test	not applicable

note: robust standard errors in parentheses
$^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$

Although the F-statistic ($6.7 < 10$) suggests potential instrument weakness, the Wu-Hausman test rejects institutional exogeneity, justifying the IV approach.

Question 6 The authors add controls for geography, climate, religion, settler identity, health, and ethnolinguistic fragmentation to show that the 2SLS coefficient on institutions is stable across specifications. Additional checks include sub-sample analyses excluding some blocks, the use of alternative instruments such as yellow fever mortality, and tests of the exclusion restriction through overidentification and by including log settler mortality in the second stage. Results are robust to alternative institutional measures, and the paper documents a historical transmission from settler mortality to early and current institutions. Their results remain overall stable, implying a significant causal effect of institutions on economic performance.

Table 9: IV regressions of Log GDP per capita

	Dependent variable:				
	Log GDP per capita				
	Base sample	Base sample	Base sample	Exclude WCA	Base sample
	(1)	(2)	(3)	(4)	(5)
Risk	0.962***	0.670***	0.690**	0.654**	0.973***
	(0.216)	(0.247)	(0.285)	(0.245)	(0.295)
Latitude	−0.417		−0.360	−0.191	
	(1.308)		(1.025)	(0.998)	
Malaria index 1994		−0.593	−0.609	−0.682	
		(0.462)	(0.459)	(0.485)	
Africa dummy					−0.471
					(0.356)
Asia dummy					−1.002**
					(0.400)
Other continent dummy					−0.920
					(0.842)
Observations	64	62	62	51	64
R^2	0.136	0.529	0.509	0.556	0.232
Adjusted R^2	0.107	0.513	0.484	0.528	0.180

Note: *p<0.1; **p<0.05; ***p<0.01

In (4) exclude contested observations in west and central Africa

Additinal controls : We add malaria index to verify that log settler mortality does not capture current health conditions. The coefficient on risk remains significant: about 0.96, and 0.67–0.69 when malaria and latitude are included. Results are robust to excluding contested west and central african observations, and adding continent dummies leaves the effect of risk large and significant (0.973), with the asia dummy negative relative to the americas.

Table 10: First stage regressions

	Dependent variable:				
	Risk				
	Base sample	Base sample	Base sample	Exclude WCA	Base sample
	(1)	(2)	(3)	(4)	(5)
Log settler mortality	-0.517^{***}	-0.426^{**}	-0.386^{**}	-0.440^{**}	-0.439^{**}
	(0.141)	(0.184)	(0.186)	(0.186)	(0.172)
Latitude	2.007		1.679	1.792	
	(1.330)		(1.393)	(1.394)	
Malaria		-0.785	-0.640	-0.795	
		(0.530)	(0.542)	(0.590)	
Africa dummy					-0.273
					(0.410)
Asia dummy					0.331
					(0.497)
Other continent dummy					1.229
					(0.840)
Observations	64	62	62	51	64
R^2	0.300	0.298	0.315	0.364	0.307
Adjusted R^2	0.277	0.274	0.280	0.323	0.260
Note:					$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$